



Ocean of Data Challenge: Foil Forward (*Featuring SailGP*)

Dates

June 3rd - June 10th, 2026

Overarching Challenge Question

How can the data generated by high-performance sailing help us better understand the sport, sharpen its competitive edge, and tell the story of wind-powered competition?

The Challenge: Foil Forward

Participants will address the ultimate question: **How can the data generated by high-performance sailing help us better understand the sport, sharpen its competitive edge, and tell the story of wind-powered competition? How can we use this data to give athletes, engineers, and fans new ways to see, predict, and celebrate what happens on the water?**

Post-secondary students across Canada are encouraged to use SailGP race telemetry, including GPS position, speed, foil rate and angle, wing rotation, and hydraulics alongside publicly available atmospheric datasets to investigate the dynamics of high-performance sailing.

During the Ocean of Data Challenge, participants will work with data at their own pace, showcase their skills, and compete for prizes including a behind-the-scenes Tech Tour with SailGP. The challenge is divided into 3 streams: **Above the Water**, **On the Water**, and **Around the Water**. Participants will select a stream and develop a solution using SailGP data.

#1 Above the Water (*Wind, atmosphere, and race performance*)

How can data about wind and atmospheric conditions help us understand, predict, and optimize race performance? How closely do pre-race forecasts match the conditions teams actually experience on the racecourse? How does disturbed air from one boat affect the performance of boats nearby? What is the relationship between wind at the top of the course and at the bottom, and how does that shape performance and fairness?

- *Forecasting outcomes:* Compare pre-race forecasts against the conditions teams actually experience on the racecourse. Where do forecasts and reality diverge, and what does that reveal about modelling wind in fast-changing coastal environments?
- *Tactical outcomes:* Visualize how disturbed air moves from one boat to another and shapes race outcomes, especially in high-density race configurations where boats are sailing in close proximity.

#2 On the Water (*Boat telemetry and performance science*)

What separates a winning run from an average one? Each F50 produces a continuous stream of telemetry across every second of every race, including GPS position, speed, foil rate and angle, wing rotation, and hydraulics. Inside that data is the signature of athlete decision-making and the patterns that turn a pack of sailors at the start line into a podium of winners. How can data science turn this into insight?

- *Performance outcomes:* Use telemetry to identify the patterns that correlate with race-leading speed. What do winning runs have in common across foil control, wing trim, and boat handling?
- *Athlete and engineer outcomes:* Build tools that help drivers, foil controllers, and wing trimmers make sharper, data-informed decisions on the water.

#3 Around the Water (*fan experience and spectator experience*)

SailGP generates more than race results — it creates an entire fan experience across the shoreline, broadcast, and digital world. From live race data and event movement to audience engagement and storytelling, there is an opportunity to help fans better understand, follow, and connect with the action on and off the water. How can data transform the spectator experience?

- *Fan experience outcomes:* Use race, event, or audience data to create more immersive, interactive, and accessible ways for fans to experience SailGP. How can data help both lifelong sailing fans and first-time spectators better understand the sport?
- *Event and entertainment outcomes:* Build tools, visualizations, or experiences that enhance engagement before, during, and after the race. This could include live insights, personalized experiences, crowd flow, storytelling, social engagement, or digital fan experiences.

The Rules

- Competitors must either own the rights to all software used in their project or have written permission from the owner.
- Submissions can range from an infographic to a machine learning model for predictions. Surprise us! Projects will be judged on their clarity, creativity, and potential impact.
- Competitors can participate **individually or in teams of up to 4 members**.
- Submissions must NOT include any confidential information.
- Each project must use at least one dataset provided by **SailGP**, with the source clearly cited in the presentation, infographic, visual, or slides. Other data sources can be used in conjunction with SailGP data.
- Competitors must currently live in Canada and have an active Canadian bank account.
- Competitors must be a current part-time or full-time student at a post-secondary institution in Canada, OR have graduated from one of these institutions within the past 12 months.

Final Submissions

- Submission Requirements:
 - In order to be eligible, every team member must complete the [registration form](#).
 - One team member is responsible for submitting the following items to **info@deepsense.ca** using the subject line "Challenge Submission by **TEAM NAME**":
 -  **SailGP Team Concept Submission Form**
 - A link to the recorded video presentation ~ Max 3 minutes in length
 - Any additional supporting materials
- The solution you present can take any format (such as an infographic, chatbot, an app, a computer vision model, or a time series forecasting model) as long as it addresses one of the challenge streams.
- We understand that solutions will vary based on the unique skills, backgrounds, and expertise of participants.

Schedule (Atlantic Time)

- **Wednesday, June 3rd | 4:00 - 6:00 pm ADT:** Challenge kickoff! Join us in-person at The PIER, 1209 Marginal Rd, with a panel featuring Leith Rankin, Director of Customer Success at Dash Social; Tim Hornsby, Technical Director and Athlete at NorthStart SailGP, and Dr. Angela Siegel, Associate Dean Academic at Dalhousie University's Faculty of Computer Science.

- **Wednesday, June 10th | 11:59 pm ADT:** Final submissions due! Ensure that your team submits the Team Concept Submission Form and final video presentation via email to info@deepsense.ca. Use the subject line: "Challenge Submission by *TEAM NAME*."
- **Saturday, June 13th:** Winners announced via social media! All participants will be contacted with the results via email.
- **Saturday, June 20th | 12:00 pm ADT:** Winners will get the opportunity to attend a behind the scenes tech-tour during the Race week at King's Wharf, Dartmouth.

Judging and Prizes

The judging process will occur over the course of 2 days. During this time, judges will review each submission and complete an evaluation form. Once all evaluations are in, winners will be determined based on the combined results. Projects will be judged on design, completeness, technical difficulty, usefulness/business potential, originality/ingenuity, and presentation quality. The breakdown of prizes is as follows:

- First place: \$1,000
- Second place: \$600
- Third place: \$400

Moreover, the winners will also get the opportunity to attend a behind-the-scenes Tech Tour of SailGP race on **Saturday, June 20th!**

Data

For this challenge, participants are required to use at least one dataset from SailGP's data.

Data link: <https://drive.google.com/file/d/1yXLn4tzXRdJ2C-udGX4OavMSsRleAblx/view>

Resource Pages

These resource pages offer additional datasets that can be used in conjunction with SailGP data.

-  **Resources Page** : This page will help you get started with the basics of SailGP
-  **Supplementary Data Sources**

Links to selected winners' presentations from recent challenges

- OOD 9: Halting the Harbour — [Second Place Presentation](#)
- OOD 10: Coastal Clarity — [First Place Presentation](#)
- OOD 12: AI in Action — [First Place Presentation](#)
- OOD 13: Our Living Ocean — [First Place Presentation](#)